ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Tsunami Monitoring with Remote Sensing

Detecting the wave at sea, warning the coast, and mapping what it leaves behind

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GOAL OF THIS LECTURE

Use satellites and ocean sensors to detect a tsunami, warn the coast in time, and map the flooding and damage it leaves.

Wave physics, the warning chain, sensors in the sea and in space, damage mapping, hazard maps, and real tsunamis from 2004 to 2025.

THIS LECTURE

Outline

01 Tsunamis and why we monitor them
Causes, wave physics, the seafloor source, India’s threat, and before–during–after

02 Detecting a tsunami at sea
The warning chain, GNSS, ocean-bottom sensors, satellite altimetry, the ionosphere and travel times

03 Mapping what it leaves behind
Optical and radar damage mapping, inundation and run-up, and land that moved

04 Preparing for the next one
Elevation models, hazard and evacuation maps, and Tsunami Ready communities

05 Case studies
Indian Ocean 2004, Tōhoku 2011, Palu 2018 and Hunga Tonga 2022

06 Engineering practice
Where engineers use tsunami data, a workflow, limits and exercises

01

PART 1

Tsunamis and why we monitor them

Causes, wave physics, the seafloor source, India’s threat, and before–during–after

TSUNAMIS

What makes a tsunami

Earthquakes

Most tsunamis. A subduction-zone earthquake lifts the seafloor over a wide area in seconds.

Landslides

Under the sea or into it. Local, but can be very high: Palu, 2018.

Volcanoes

Explosions, collapses and pressure waves: Hunga Tonga, 2022.

Weather

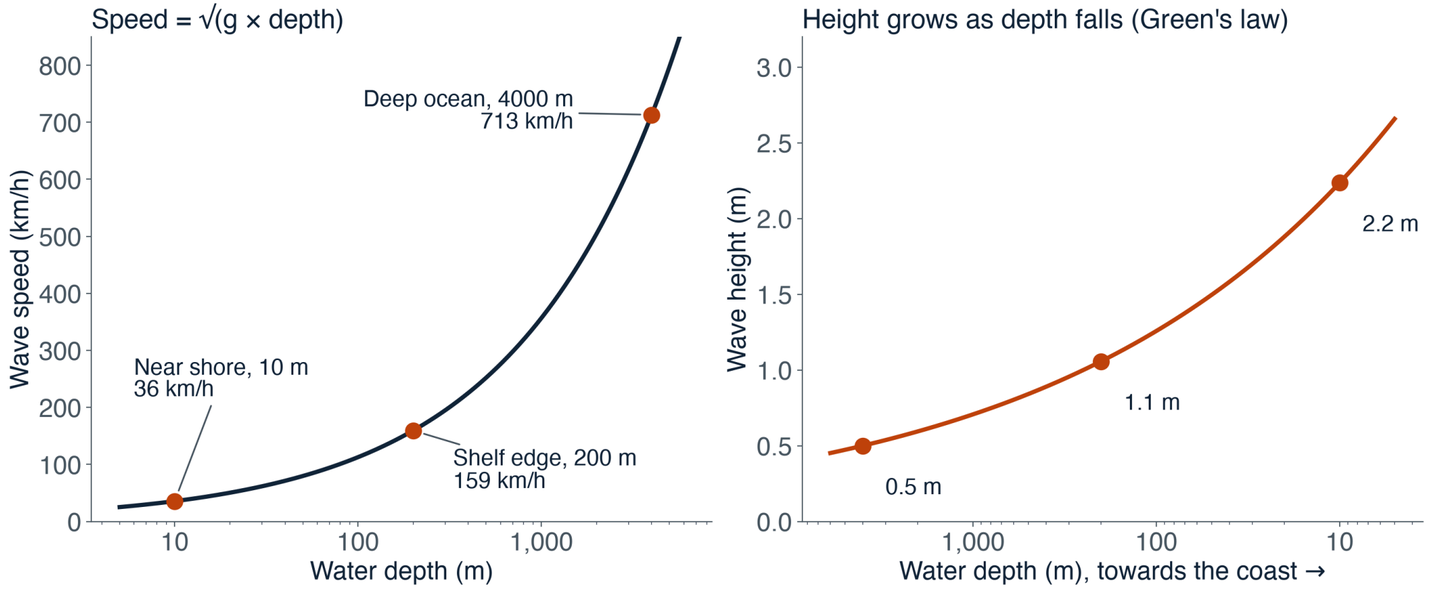
Meteotsunamis: a fast-moving jump in air pressure pushes a long wave.

Not a wind wave

A wind wave moves the top few metres of water. A tsunami moves the whole water column, with wavelengths of hundreds of kilometres and periods of 10 minutes to over an hour.

TSUNAMIS

Fast in deep water, high near the coast



Computed from long-wave theory and Green’s law.

$$c = \sqrt{(g \times h)}$$

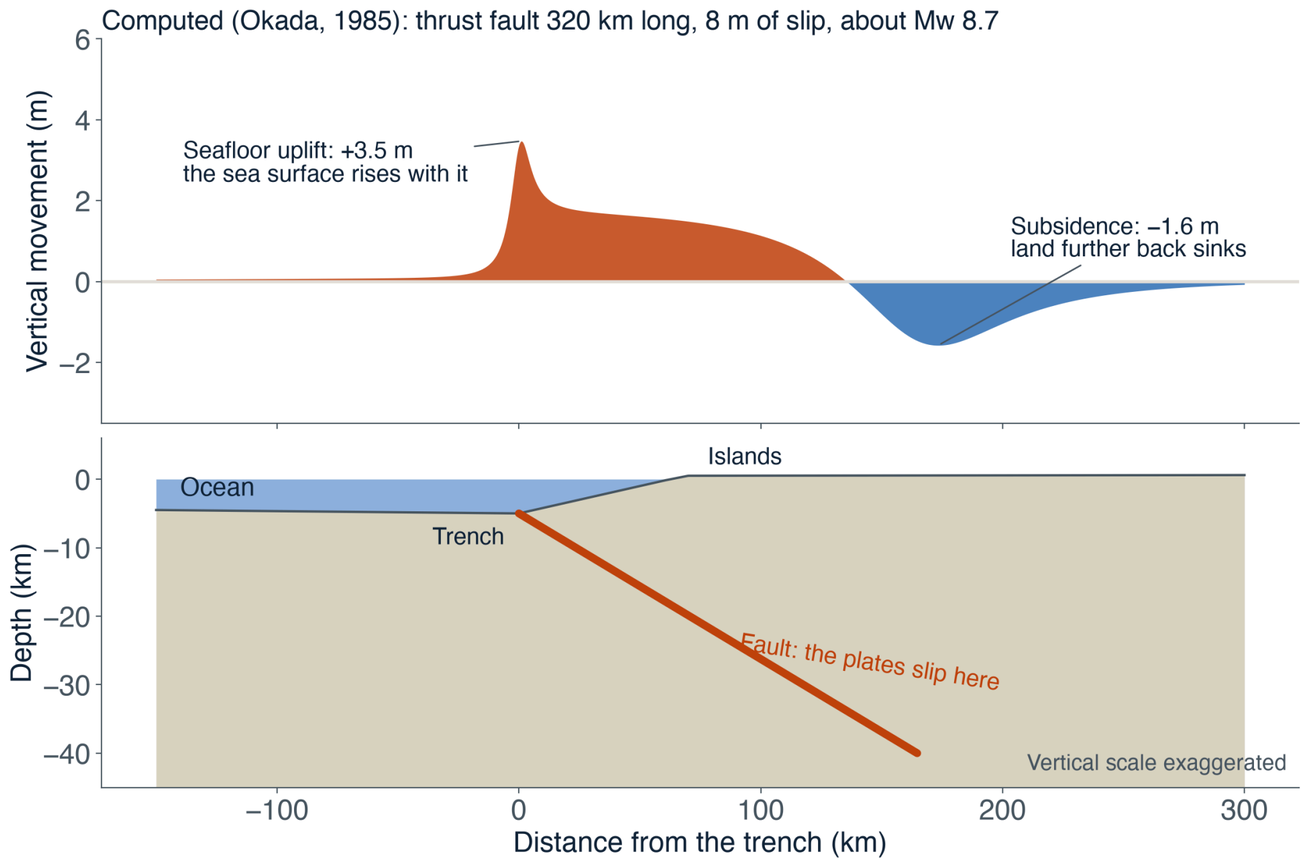
Deep ocean: about 713 km/h, as fast as a jet, but often under a metre high.

Near the shore it slows to about 36 km/h and the back of the wave catches up.

Height grows as depth falls: 0.5 m becomes about 2.2 m before it even breaks.

TSUNAMIS

The seafloor lifts the sea



Computed: Okada (1985), about Mw 8.7.

A megathrust earthquake lifts a wide strip of seafloor in under a minute; the sea surface copies it.

Uplift on one side, subsidence on the other: one coast sees a crest first, the other sees the sea draw back.

The same model fits InSAR and GNSS on land (Lecture 5), and its uplift starts the tsunami models.

TSUNAMIS

India’s tsunami threat

10,749
Deaths in India in 2004, with 5,640 missing

≈ 230,000
Deaths around the Indian Ocean, 14 countries

1945
Makran tsunami: over 4,000 deaths; 2 m waves at Mumbai

East coast and islands

The Andaman–Sumatra subduction zone. The islands get minutes; the mainland coast about two hours.

West coast

The Makran subduction zone off Pakistan and Iran threatens Gujarat, Mumbai and the west coast.

A warning system since 2007

INCOIS, Hyderabad, runs India’s Tsunami Early Warning Centre, around the clock.

Sources: Government of India (2004 figures); USGS; UNESCO-IOC; INCOIS.

TSUNAMIS

Before, during and after

Stage	Question	Remote sensing and sensors
Before	Who is exposed, and where should they go?	Elevation models, bathymetry, land use; hazard and evacuation maps
First minutes	Can this earthquake make a tsunami?	Seismometers, GNSS, pre-computed scenarios
Wave at sea	How big is it, and where is it going?	Ocean-bottom pressure sensors, tide gauges, altimeters, the ionosphere
After	What was flooded and damaged?	Optical and radar images, InSAR, drones, field surveys

02

PART 2

Detecting a tsunami at sea

The warning chain, GNSS, ocean-bottom sensors, satellite altimetry, the ionosphere and travel times

DETECTION

The warning chain

0 min
Earthquake

≈ 3–5 min
Location and size from seismometers and GNSS

≈ 10–15 min
First bulletin from pre-computed scenarios

30–60 min
Ocean sensors confirm the wave

Hours
Updates, then the all-clear

A warning is first issued from the earthquake alone, by matching it to thousands of tsunamis modelled in advance. Sensors in the sea then confirm, upgrade or cancel it.

India

INCOIS monitors seismic stations, ocean-bottom pressure recorders and tide gauges across the Indian Ocean, 24 hours a day.

DETECTION

Measuring a great earthquake quickly

M 7.9

Japan’s first estimate for Tōhoku, 3 minutes after the shaking began

Mw 9.0

The real size: about 45 times more energy

The first warnings, of 3 to 6 m, were far too low; run-up reached 40.5 m.

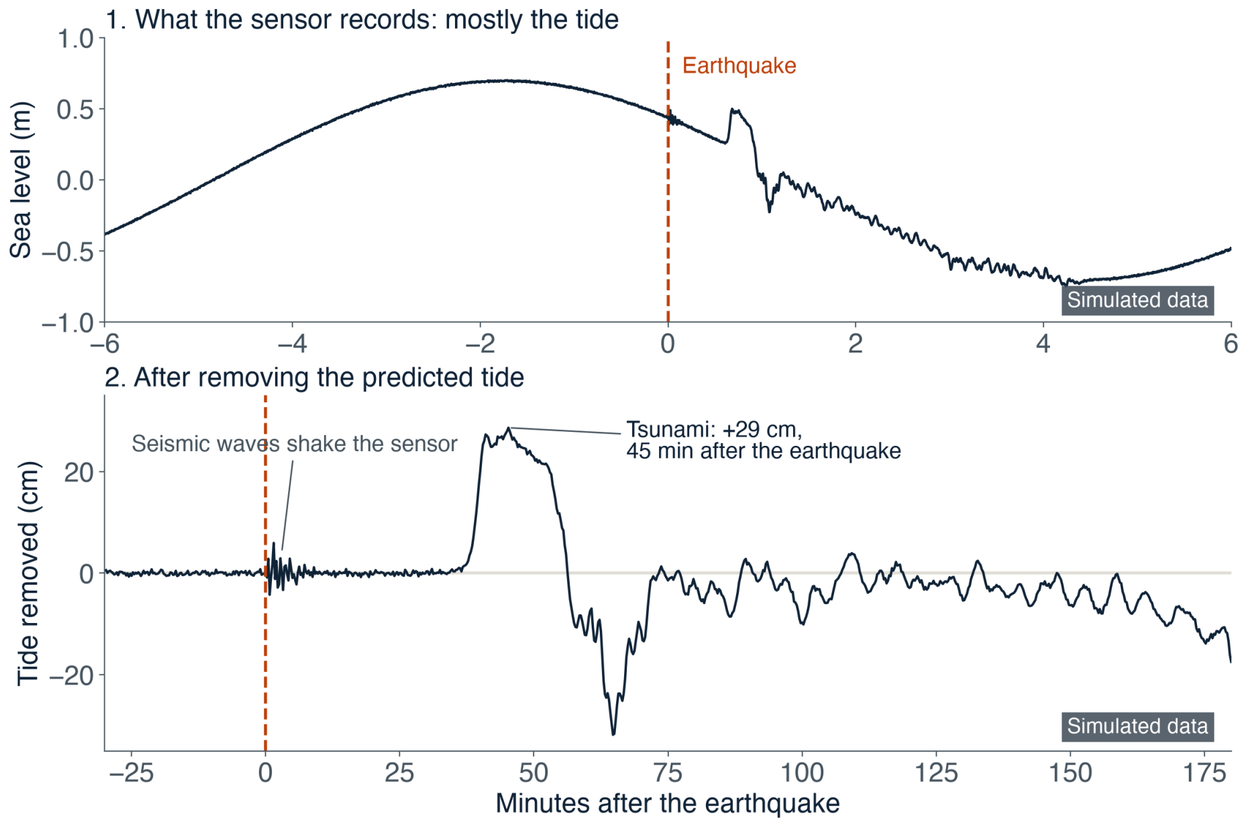
Seismic magnitude estimates saturate for the very largest earthquakes.

GNSS stations measure how far the ground actually moved, which shows the true size within minutes.

GPS buoys 10 to 20 km offshore saw the sea rising; Japan raised its warnings at 15:14 and 15:30.

DETECTION

Sensors on the sea floor



Simulated: a sensor in 4000 m of water.

A pressure sensor on the sea floor feels the weight of the water above it.

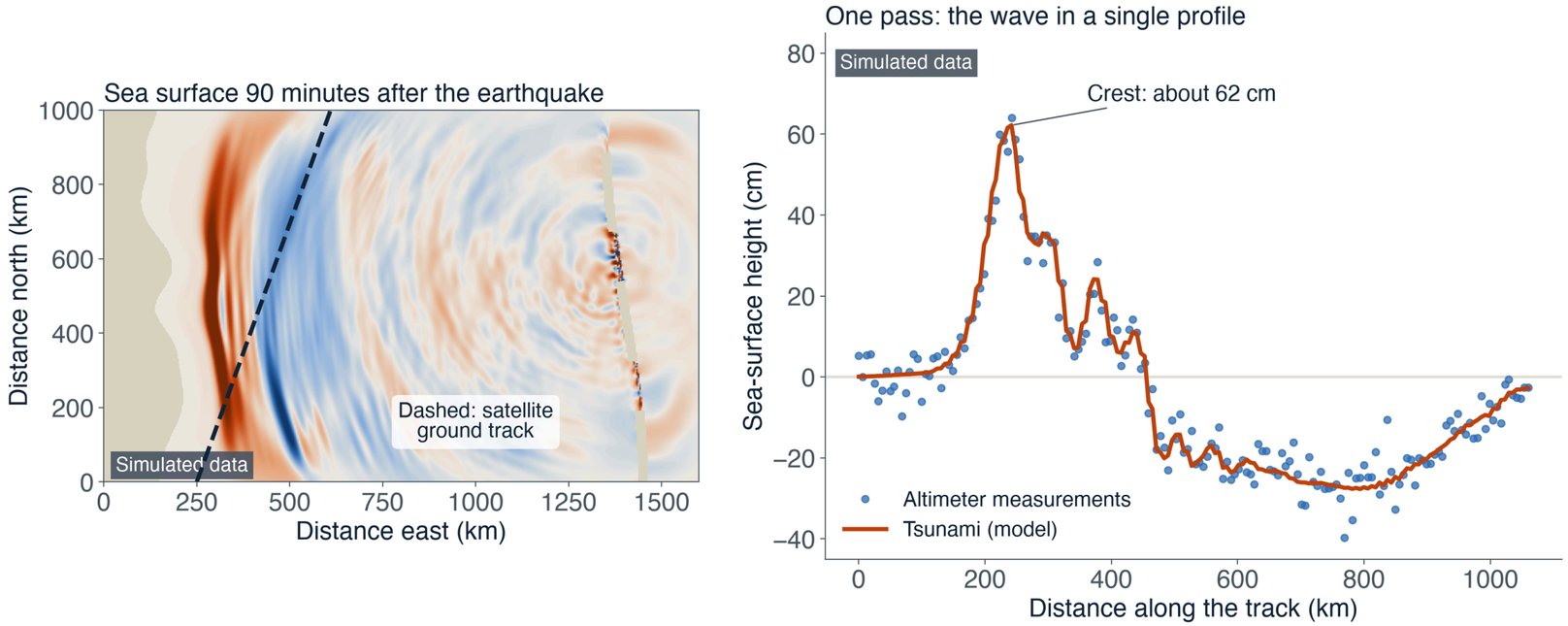
It can detect a tsunami 1 cm high in 6000 m of water, once the tide is removed.

An acoustic link sends data to a surface buoy, then by satellite to the warning centre.

NOAA’s DART network covers the Pacific; India runs its own sensors in the Bay of Bengal and Arabian Sea.

DETECTION

Satellite altimeters



Simulated: one altimeter pass across the wave, crest about 62 cm.

26 December 2004: Jason-1 and TOPEX/Poseidon crossed the Indian Ocean about 2 hours after the earthquake and measured a wave about 60 cm high: the first tsunami clearly seen from space.

Luck, not warning

An altimeter sees only a line below it; a pass over a tsunami is rare.

DETECTION

SWOT: a tsunami in two dimensions

Mw 8.8
Kamchatka earthquake, 29–30 July 2025

70 min
After the earthquake, SWOT crossed the wave

> 45 cm
Height of the leading wave in the open ocean

120 km
Width of SWOT’s radar swath

A wide swath shows the wave’s shape and direction, not just a single line.

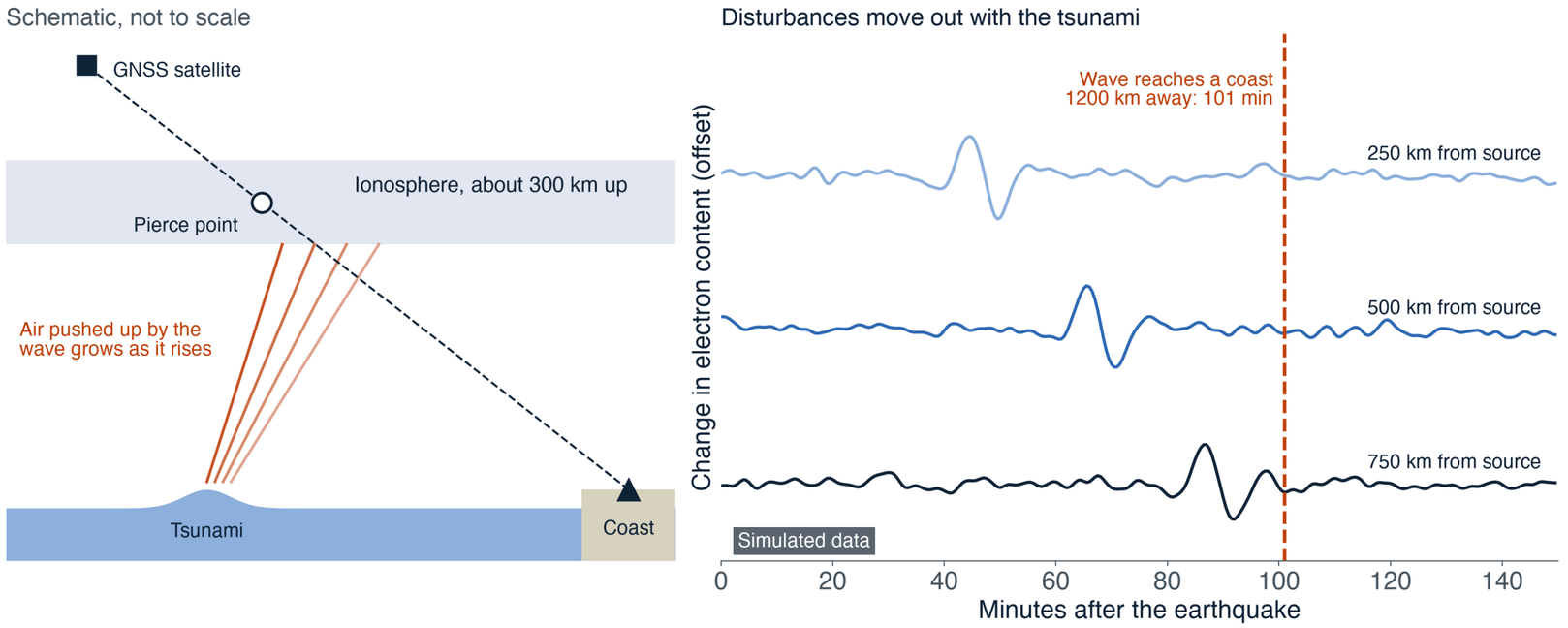
SWOT also saw smaller waves trailing the front, which point to slip close to the trench.

The data test the forecast models that warning centres use.

Sources: NASA JPL (2025); Sepúlveda et al. (2026), Science; Ruiz-Angulo et al. (2025), The Seismic Record.

DETECTION

Seeing a tsunami in the sky



Left: schematic. Right: simulated.

NASA’s experimental **GUARDIAN** system uses GNSS stations around the world. In 2025 it flagged the Kamchatka tsunami in as little as 20 minutes, 30 to 40 minutes before the wave reached Hawai’i.

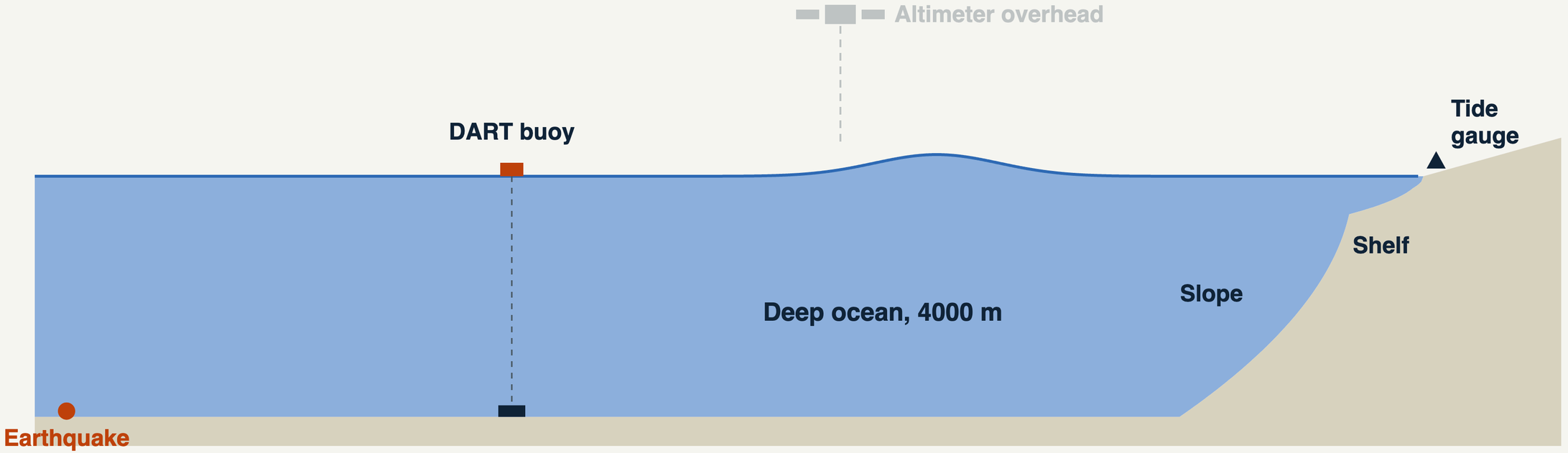
Why it matters

It needs no buoys: GNSS stations already on land watch the sky over open ocean.

DETECTION

A tsunami crossing the ocean

Computed: speed = √(g × depth); height from Green's law. Wave heights exaggerated; depths on a square-root scale.

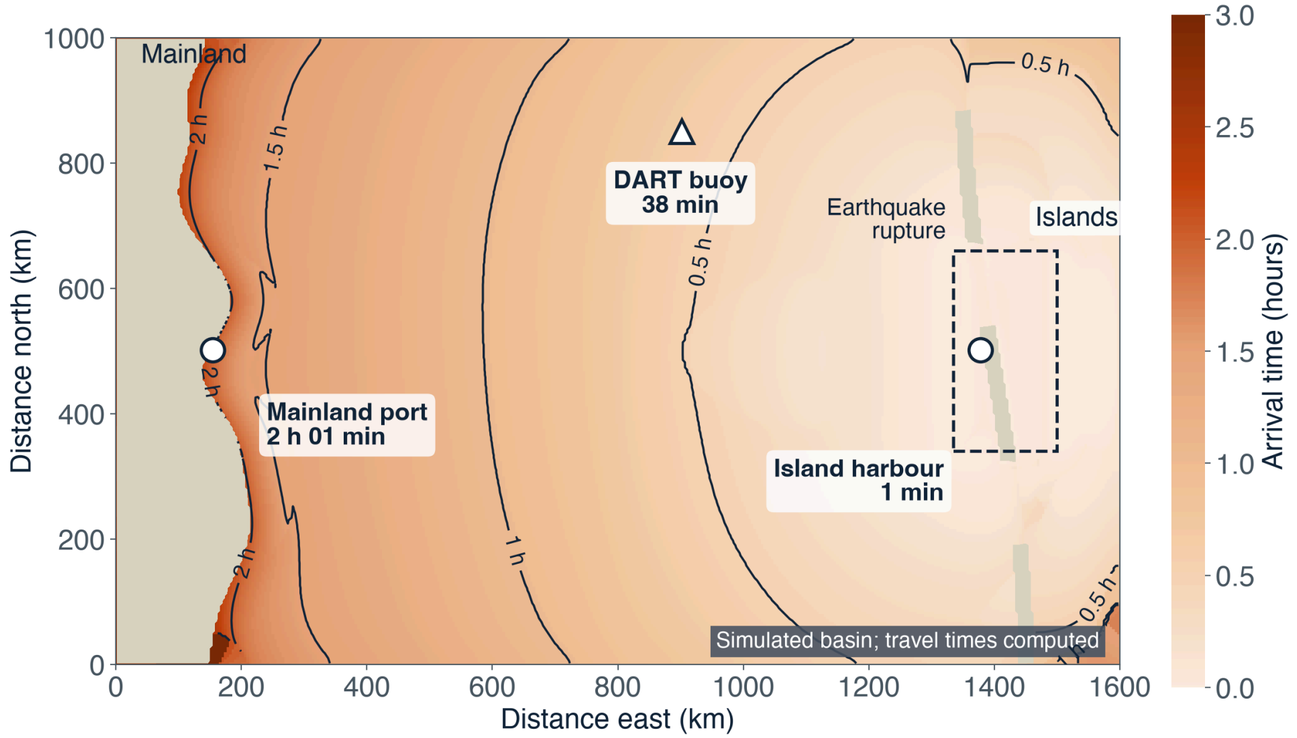


1 h 12 min Depth under the crest: 4000 m Speed: 713 km/h **Height: 0.5 m**

DART buoy detects it: 0 h 38 min **Satellite measures it: 1 h 04 min** **Reaches the tide gauge: waiting**

DETECTION

How much time is there?



Computed with the long-wave equations on a simulated basin.

Island harbour next to the rupture: about 1 min. Shaking is the only warning.

Mainland port about 1,200 km away: about 2 h 01 min. Enough time to warn and evacuate.

Shallow water slows the wave: the contours crowd together near the coast.

2004: Port Blair about 50 minutes; Chennai about 2½ hours.

03

PART 3

Mapping what it leaves behind

Optical and radar damage mapping, inundation and run-up, and land that moved

AFTER THE WAVE

Optical images: before and after

Very high resolution

Satellites with pixels under 1 m, such as Pléiades, WorldView and Cartosat, show single buildings, debris lines and broken roads.

Water and vegetation

Water and vegetation indices, as in flood and drought mapping, show where the sea went and where plants died.

The inundation line

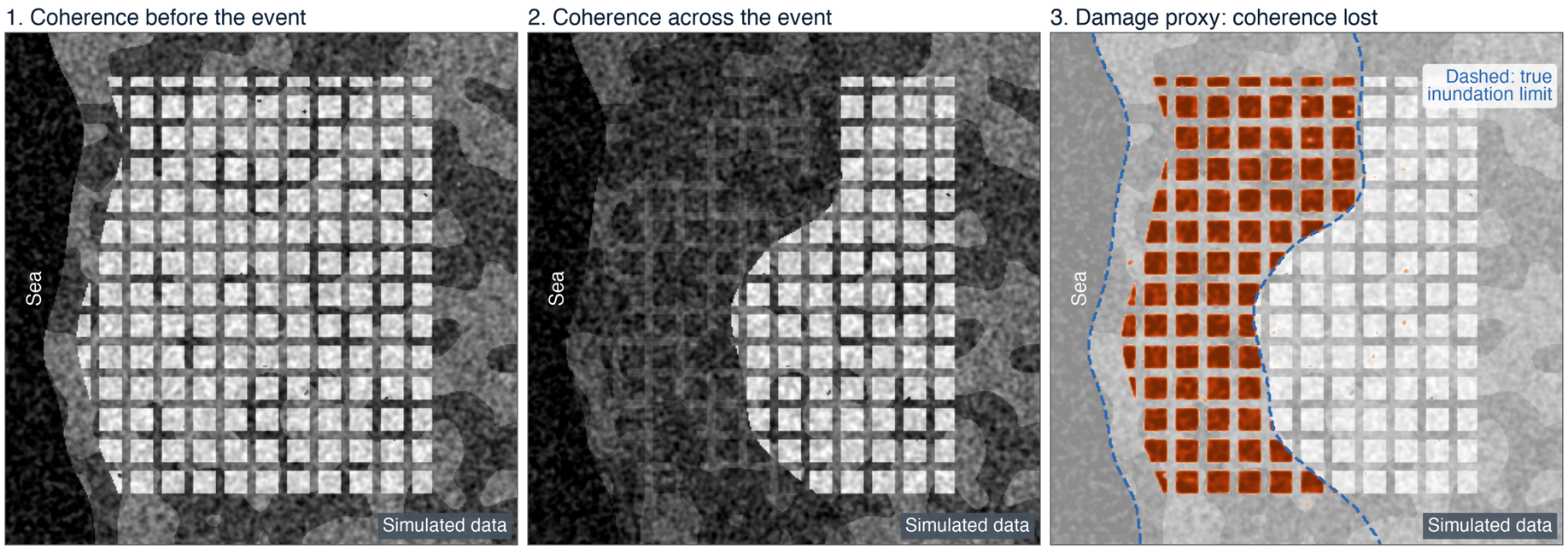
Salt-killed crops and mangroves, and lines of debris, mark the inundation limit for weeks.

Fast access

The International Charter: Space and Major Disasters gives free images to responders within days.

AFTER THE WAVE

Radar damage maps



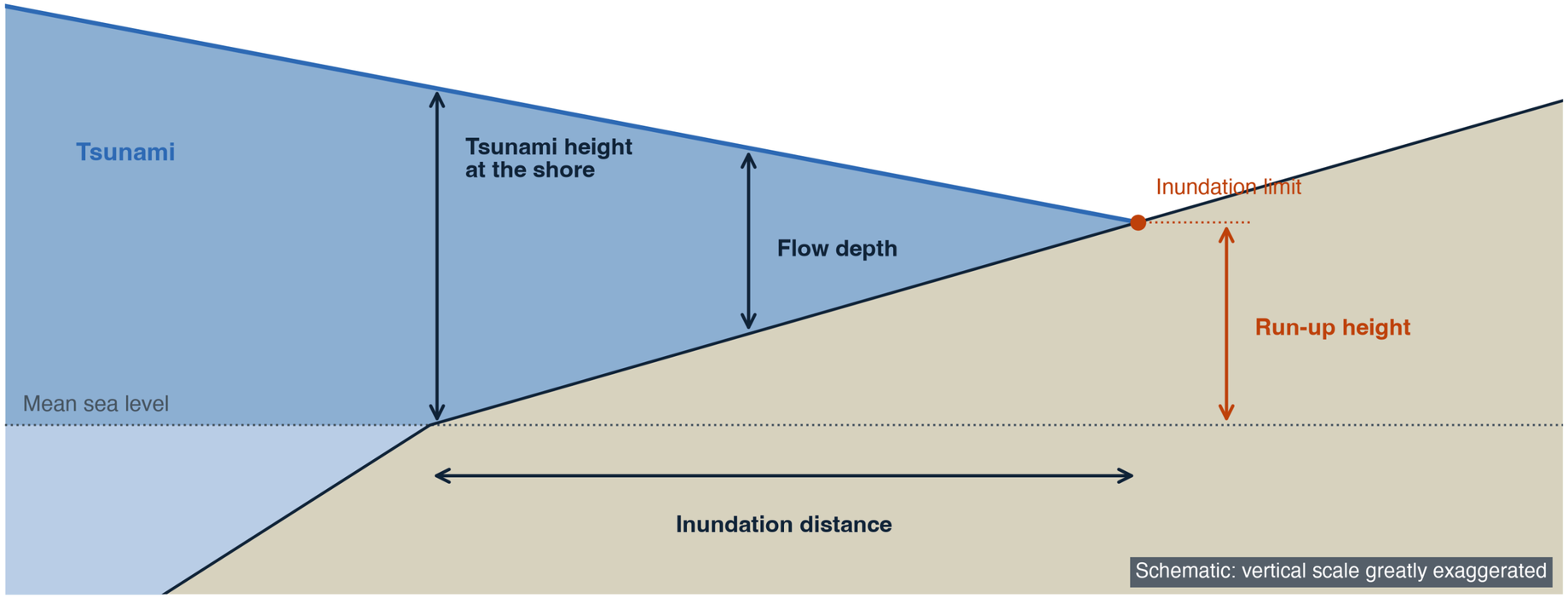
Simulated: a coastal town about 3.6 km across, 10 m pixels.

Coherence measures whether the surface stayed the same between two radar images. Buildings normally keep it; buildings swept away lose it.

NASA–JPL's ARIA team made damage proxy maps like this for Palu in 2018 from ALOS-2 radar data, through cloud and at night.

AFTER THE WAVE

Measuring how far and how high



Inundation distance: how far inland the water reached.

Run-up height: the ground height at that limit, above sea level.

Flow depth: water over the ground; it sets the forces on buildings.

AFTER THE WAVE

Land that moved

2004, Andaman and Nicobar

Parts of the Andaman Islands rose by about a metre; the Nicobar Islands sank, leaving coasts and jetties under water.

2011, Tōhoku

GNSS showed the coast had sunk by up to about 1.2 m, so high tides now reached land that had been dry.

Measuring it

InSAR and GNSS map the change, as for any earthquake (Lecture 5).

Why it matters

Sea walls, ports and drains built for the old ground level no longer work. Elevation maps must be updated.

04

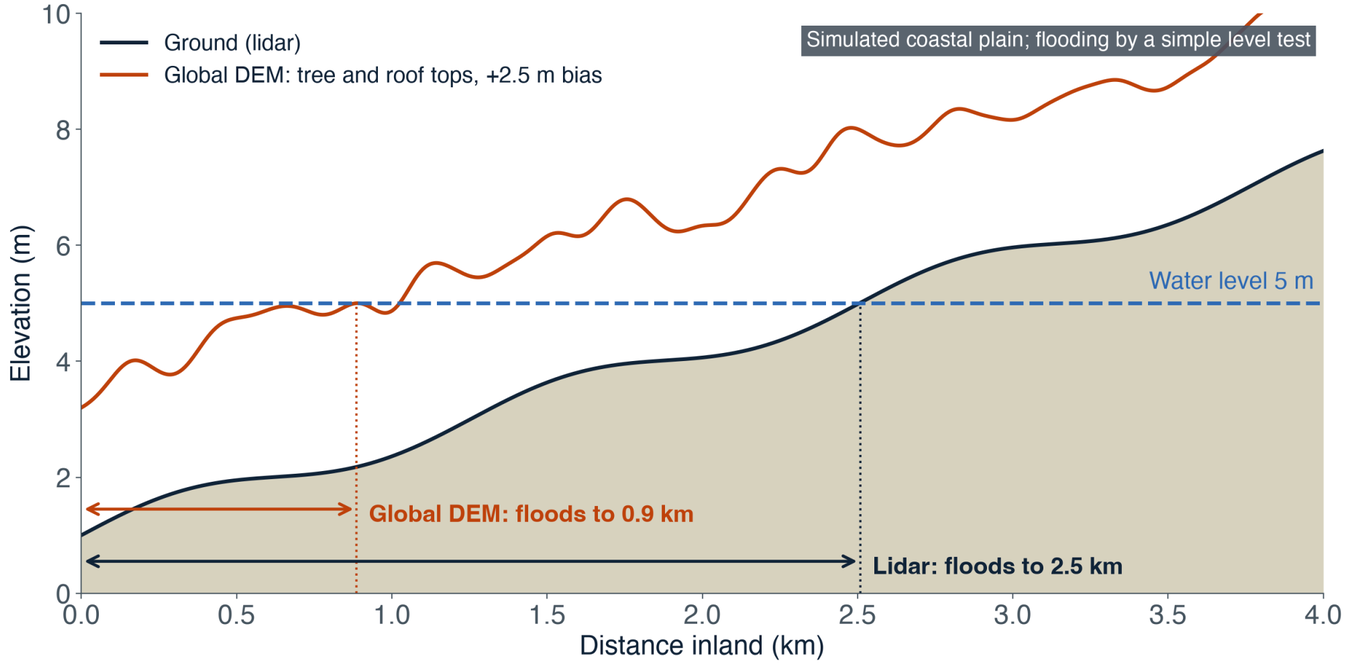
PART 4

Preparing for the next one

Elevation models, hazard and evacuation maps, and Tsunami Ready communities

PREPARING

Elevation is everything



Simulated plain; flooding by a simple level test.

Inundation maps need ground height accurate to a metre or better near the coast.

Global DEMs such as SRTM and Copernicus measure tree and roof tops, with errors of metres.

Here a 2.5 m bias shrinks the flooded strip from 2.5 km to 0.9 km.

Use lidar, drone surveys or corrected coastal DEMs for hazard maps.

PREPARING

Hazard maps and evacuation

Scenarios

Model many possible earthquakes and map the worst credible flooding.

Evacuation

Routes, safe areas and tall buildings for shelter, from elevation and building maps.

Time

Compare walking time to safety with the wave’s arrival time.

Tsunami Ready in Odisha

Venkatraipur (Ganjam) and Noliasahi (Jagatsinghpur) were the first UNESCO-IOC Tsunami Ready communities in the Indian Ocean, in 2020.

By December 2024, 26 villages in Odisha had the recognition: hazard maps, evacuation routes, signs, drills and a local plan.

Sources: UNESCO-IOC; INCOIS; Odisha State Disaster Management Authority.

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PART 5

Case studies

Indian Ocean 2004, Tōhoku 2011, Palu 2018 and Hunga Tonga 2022

CASE STUDY

Indian Ocean, 26 December 2004

Mw 9.1
Sumatra–Andaman earthquake: 1,300 to 1,600 km of fault

≈ 230,000
Deaths in 14 countries

≈ 2½ h
Until the wave reached Chennai; about 50 minutes at Port Blair

None
Tsunami warning system in the Indian Ocean at the time

Jason-1 and TOPEX/Poseidon measured the wave from space: about 60 cm, 2 hours after.

Before-and-after satellite images mapped damage along every coast.

Led to India’s warning centre (2007) and the Indian Ocean warning system.

Sources: USGS; Government of India; EERI (2005); Ablain et al. (2006).

CASE STUDY

Tōhoku, Japan, 11 March 2011

Mw 9.0
Off the Pacific coast of Tōhoku

40.5 m
Highest run-up, at Aneyoshi, Miyako

561 km²
Land flooded, mapped from aerial photos and satellite images

≈ 18,400
Dead (15,900) and missing (2,520)

The first warning, from a magnitude of 7.9, was far too low; GPS buoys led to upgrades.

Radar and optical images mapped flooding and damage within days.

The Fukushima Daiichi nuclear plant flooded: a lesson for critical facilities.

Sources: JMA; Ozaki (2011); Mori et al. (2012); Japan National Police Agency; GSI.

CASE STUDY

Palu, Sulawesi, 28 September 2018

Mw 7.5
A strike-slip earthquake: not expected to make a large tsunami

> 4,000
Deaths from the earthquake, tsunami and liquefaction

Minutes
Waves reached Palu almost at once, inside a narrow bay

ALOS-2
Radar used for NASA–JPL damage proxy maps

Landslides in the bay probably added to the tsunami: very hard to warn for.

InSAR mapped a rupture about 170 km long right through Palu.

Lesson: for local tsunamis, shaking is the warning: go to high ground at once.

Sources: NASA JPL ARIA; Carvajal et al. (2019); field surveys in Pure and Applied Geophysics (2021).

CASE STUDY

Hunga Tonga–Hunga Ha’apai, 15 January 2022

58 km
Height of the eruption plume: a record

≈ 15 m
Run-up on Tongatapu, within about 10 minutes

10,000 km
Away in Peru, two people drowned

Global
The pressure wave crossed every ocean

The pressure wave from the explosion travelled faster than a normal tsunami and drove waves earlier than forecast.

Weather satellites (GOES-17, Himawari-8) imaged the plume and the expanding pressure wave.

Lesson: systems built for earthquakes can miss volcanic and atmospheric tsunamis.

Sources: NOAA NCEI; Omira et al. (2022), Nature; UNESCO-IOC.

06

PART 6

Engineering practice

Where engineers use tsunami data, a workflow, limits and exercises

ENGINEERING

Where engineers use tsunami data

Critical facilities

Place power, water and nuclear plants above the worst credible flooding, or protect them.

Ports and harbours

Strong currents and surges inside harbours damage ships, moorings and breakwaters.

Vertical evacuation

Strong buildings and towers to climb when there is no time to leave.

Design loads

Flow depth and speed set the forces on buildings: ASCE 7 (2016) added a tsunami chapter.

Lifelines

Coastal roads, railways, bridges, pipelines and power lines.

Land-use planning

Keep schools and hospitals out of the worst zones; keep mangrove and forest belts.

ENGINEERING

A tsunami hazard workflow

01
Identify sources: faults, landslides, volcanoes

02
Get bathymetry and a good DEM

03
Model scenarios and travel times

04
Map inundation and exposure

05
Plan warnings and evacuation

06
Monitor: seismic, GNSS, ocean, satellites

07
After an event: map damage

08
Update DEMs, maps and plans

Free data and tools: GEBCO bathymetry, Copernicus DEM, Sentinel-1 and -2, and the GeoClaw and COMCOT tsunami models.

Student project: travel times from the Andaman and Makran zones to your nearest coast.

ENGINEERING

Limits and good practice

Limits

- Altimeter passes over a tsunami are rare
- Optical images need clear skies
- Global DEMs are too rough near the coast
- Local tsunamis arrive before any sensor can help

Good practice

- Combine seismic, GNSS and ocean sensors
- Use lidar or corrected DEMs for maps
- Teach: shaking or a falling sea means go
- Check maps against surveys and past events

EXERCISES

Try these

01 Find the speed of a tsunami in 4000 m and in 50 m of water, in m/s and km/h.

02 A tsunami is 0.4 m high in 4000 m of water. Using Green’s law, how high is it in 10 m of water?

03 A tsunami starts 1,300 km away across ocean 4000 m deep. Roughly when does it arrive? If the first warning takes 15 minutes, how much time is left?

04 After a local earthquake the wave arrives in 25 minutes. People leave after 5 minutes and walk at 1 m/s. How far can they get? Is a hill 1.5 km inland safe enough?

EXERCISES

Answers

01 $\sqrt{(9.81 \times 4000)} \approx$ **198 m/s ≈ 713 km/h**. $\sqrt{(9.81 \times 50)} \approx$ **22 m/s ≈ 80 km/h**.

02 $0.4 \times (4000 \div 10)^{1/4} = 0.4 \times 4.47 \approx$ **1.8 m**, before breaking and run-up.

03 1,300,000 ÷ 198 ≈ 6,570 s ≈ **1 h 50 min**; about **1 h 35 min** left after the warning. Shallow water near the coast adds a little more time.

04 (25 – 5) × 60 × 1 = **1,200 m**. The hill at 1.5 km is out of reach: a closer high place or a tall, strong building is needed.

RECAP

Key points

01 A tsunami moves the whole ocean: fast and low at sea, slow and high at the coast.

02 Warnings start from the earthquake; GNSS and ocean sensors confirm the wave.

03 Altimeters, SWOT and the ionosphere now see tsunamis from space.

04 After the wave, optical and radar images map flooding and damage.

05 Good elevation data and evacuation maps save the most lives.

FURTHER READING

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Thank you. Questions?

Figures and the animation were made for this lecture. Wave speed and height, the seafloor uplift (Okada, 1985) and the animation are computed; the ocean basin is simulated and the tsunami across it computed with the long-wave equations; the sensor records, altimeter profile, ionosphere signals, radar scene and elevation profile are simulated and labelled as such; two diagrams are schematic. Case-study facts: USGS, Government of India, EERI, JMA, NASA JPL, NOAA, UNESCO-IOC, INCOIS and the papers on the previous slide.